

RATES OF CONVERGENCE OF LINEAR COMBINATIONS OF ORDER STATISTICS¹

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1. Introduction

Let $X_{1n}, X_{2n}, \dots, X_{nn}$ be the order statistics of an independent sample of size n from a population having a continuous distribution function F . In this paper we consider the rates of convergence to normality of statistics, suitably normalized, of the form

$$(1) \quad T_n = n^{-1} \sum_{j=1}^n c_{jn} h(X_{jn}),$$

where $\{c_{jn}\}$ are specified real numbers and h is a measurable function on the real line.

As in Chernoff, Gatsworth and Johns (1967) T_n can be decomposed as

$$(2) \quad T_n = \mu_n + n^{-1/2} Q_n + n^{-1/2} R_n,$$

where μ_n is non-random, Q_n is effectively a linear combination of independent random variables, and R_n is a remainder term. The idea is to find conditions under which a rate can be specified for the convergence of Q_n to normality and R_n to zero.

In section 2 we derive the decomposition (2) and give a number of results needed in the sequel. In section 3 we present the main results and in section 4 we consider the case where $\{c_{jn}\}$ are zero for j/n outside a fixed interval bounded away from 0 and 1.

A similar study has been reported in Rosenkrantz and O'Reilly (1972). Their approach, however, is completely different from the present one. The best possible rate they can obtain is $n^{-1/4} \log n$. Although their conditions differ from ours this is not as good as our best possible rate, which is $n^{-1/2} (\log n)^2$.

2. Decomposition of T_n and preliminary results

Suppose F is strictly increasing so that F^{-1} is uniquely defined and set $H(x) = h(F^{-1}(x))$. In the rest of this paper we assume that H' exists on $(0,1)$.

Denoting by $U_{in} = F(X_{in})$, $i=1, \dots, n$, the uniform $(0,1)$ order statistics and applying Taylor's theorem, we get

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$$\begin{aligned}
(3) \quad T_n &= n^{-1} \sum_j c_{jn} H(U_{jn}) \\
&= n^{-1} \sum_j c_{jn} H(j/(n+1)) \\
&\quad + n^{-1} \sum_j c_{jn} V_{jn} H'(j/(n+1)) \\
&\quad + n^{-1} \sum_j c_{jn} V_{jn} (H'(U_{jn}^*) - H'(j/(n+1))) \\
&= \mu_n + n^{-1/2} Q_n + n^{-1/2} R_n \quad (\text{say})
\end{aligned}$$

where $V_{jn} = U_{jn} - j/(n+1)$ and U_{jn}^* lies between U_{jn} and $j/(n+1)$.

It is well known (see, e.g., Breiman (1968), p. 285) that the joint distribution of $\{U_{jn}; j=1, \dots, n\}$ is the same as that of $\{S_j/S_{n+1}; j=1, \dots, n\}$ where $S_j = Z_1^* + \dots + Z_j^*$ with $\{Z_i^*\}$ independent random variables each exponentially distributed with parameter one. Put $Z_i = Z_i^* - 1$ and let $\psi(x, y)$ be the function

$$(4) \quad \psi(x, y) = \begin{cases} 1 - y & \text{for } x \leq y \\ -y & \text{for } x > y \end{cases}$$

Using the above representation, it follows that

$$(5) \quad V_{jn} = S_{n+1}^{-1} \sum_{i=1}^{n+1} \psi_{ijn} Z_i$$

with $\psi_{ijn} \equiv \psi(i/(n+1), j/(n+1))$. This representation of V_{jn} differs from the one in Chernoff, Gastwirth and Johns (1967). We find the present one more convenient to work with.

Using (5), Q_n becomes

$$(6) \quad Q_n = n S_{n+1}^{-1} n^{-1/2} \sum_i \alpha_{in} Z_i,$$

where $H'_{jn} \equiv H'(j/(n+1))$ and

$$(7) \quad \alpha_{in} = n^{-1} \sum_j c_{jn} H'_{jn} \psi_{ijn}.$$

Since $n S_{n+1}^{-1} \rightarrow 1$ w.p.1, we write

$$(8) \quad Q_n = L_n + R_{1n}$$

with $L_n = n^{-1/2} \sum_j \alpha_{jn} Z_j$ and $R_{1n} = (n S_{n+1}^{-1} - 1) L_n$ and expect R_{1n} to tend to 0.

Note that

$$(9) \quad E L_n = 0 \text{ and } \text{Var } L_n = n^{-1} \sum_j \alpha_{jn}^2 = \sigma_n^2 \quad (\text{say}).$$

In the next section we give conditions for convergence of L_n/σ_n to normality and R_{1n}/σ_n to zero. Before proceeding with this, we give a few results.

Lemma 1

If X and Y are random variables and a, b , and c are real numbers, then

- (i) $P(X + Y \leq a + b) \leq P(X \leq a) + P(Y \leq b)$
- (ii) $P(X + Y \leq a + b) \geq P(X \leq a) - P(Y > b)$
- (iii) $P(X \geq aY/b) \leq P(X \geq a(1-c)) + P(|Y - b| \geq bc)$ for $a, b > 0$
- (iv) $P(|XY| \geq |ab|) \leq P(|X| \geq |a|) + P(|Y| \geq |b|)$.

Proof:

It follows in a straightforward manner.

Lemma 2

With S_n as before, we have

$$(10) \quad P(n^{1/2} |n S_n^{-1} - 1| \geq 2(\log n)^{1/2}) = O(n^{-1/2}).$$

Proof:

Let $\alpha_n = ((\log n)/n)^{1/2}$. Then, using Lemma 1 (iii)

$$\begin{aligned} P(n^{1/2} |n S_n^{-1} - 1| \geq 2(\log n)^{1/2}) &= P(n^{1/2} |S_n - n| \geq 2(\log n)^{1/2} S_n) \\ &\leq P(|S_n - n| \geq 2n^{1/2}(\log n)^{1/2}(1 - \alpha_n)) + P(|S_n - n| \geq n\alpha_n) \\ &\leq 2 P(|S_n - n| \geq (n \log n)^{1/2}) \end{aligned}$$

for n large enough since in that case $1 - \alpha_n \geq 1/2$. Since S_n has a gamma distribution, the result follows by applying Bernstein's inequality to the above expression.

We give one final result.

Lemma 3

$$P(n^{1/2} \max_{1 \leq j \leq n} |V_{jn}| \geq \epsilon \log n) = O(n^{-1/2}) \text{ for any } \epsilon > 0.$$

Proof:

It follows directly from Lemma 2.2 of Rosenkrantz, and O'Reilly (1972).

3. Main results

In this section we consider conditions for $n^{1/2}(T_n - \mu_n)$ to converge at a specified rate. We do this by looking at each of the terms L_n , R_{1n} and R_n .

3.1 Convergence of L_n

For the convergence of L_n we apply the Berry-Esseen theorem. (See, e.g. Chung (1968), p. 206). To this end, set

$$Z_{jn} = n^{-1/2} \alpha_{jn} Z_j / \sigma_n; \beta = E|Z_1|^3$$

and

$$(11) \quad \Gamma_n \equiv \sum_{j=1}^{n+1} E|Z_{jn}|^3 = \beta \sigma_n^{-3} n^{-3/2} \sum_j |\alpha_{jn}|^3.$$

It follows that

$$E Z_{jn} = 0; E Z_{jn}^2 = n^{-1} \alpha_{jn}^2 / \sigma_n^2$$

and thus

$$\sum_{j=1}^n E Z_{jn}^2 = 1.$$

In this case the Berry-Esseen theorem can be stated as follows.

Lemma 4.

If $\Gamma_n = o(1)$ as $n \rightarrow \infty$, then there exists a universal constant A_0 such that

$$(12) \quad \sup_x |P(L_n \leq \sigma_n x) - \Phi(x)| \leq A_0 \Gamma_n$$

To apply this we have to specify a rate of convergence for Γ_n . Let $0 < \alpha \leq 1$; to make it compatible with the rates of the other terms, we want $\Gamma_n = O(n^{-\alpha/2}(\log n)^2)$. This means

$$(13) \quad n^{-1} \sum_j |\alpha_{jn}^3| = O(n^{(1-\alpha)/2} \sigma_n^3 (\log n)^2).$$

An upper bound for the L.H.S. of (13) that should be useful is

$$(14) \quad n^{-1} \sum_j |\alpha_{jn}|^3 \leq 6n^{-3} \sum_{j,k,\ell} |c_{jn} c_{kn} c_{\ell n} H'_{jn} H'_{kn} H'_{\ell n} T_{jk\ell n}| = \Gamma_n^* \quad (\text{say})$$

where $T_{jkl n} = T(j/n + 1, k/n + 1, l/n + 1)$ and

$$(15) \quad T(x, y, z) = \min(x, y, z)(1 - \max(x, y, z)) \text{ for } 0 \leq x, y, z \leq 1.$$

A sufficient condition for (13) is

$$(16) \quad \Gamma_n^* = O(n^{(1-\alpha)/2} \sigma_n^3 (\log n)^2).$$

Remarks

1. In some cases the integral

$$(17) \quad \int_0^1 \int_0^1 \int_0^1 c(x)c(y)c(z)H'(x)H'(y)H'(z)T(x, y, z)dx dy dz$$

converges absolutely, where in this case $c_{jn} \equiv c(j/(n+1))$ for some function c . In such a case we expect (16) to hold.

2. Another upper bound for the L.H.S. of (13) is

$$\max_{1 \leq j \leq n} |\alpha_{jn}| n^{-1} \sum_j \alpha_{jn}^2 = \sigma_n^2 \max_{1 \leq j \leq n} |\alpha_{jn}|.$$

Thus a sufficient condition for (13) to hold is

$$(18) \quad \sigma_n^{-1} n^{-1/2} \max_{1 \leq j \leq n} |\alpha_{jn}| = O(n^{-\alpha/2} (\log n)^2).$$

Note that a necessary and sufficient condition for the convergence of L_n/σ_n to normality is that the L.H.S. of (18) is $O(1)$. (See, e.g. Chernoff, Gatsworth and Johns (1967).

3.2 Convergence of R_{1n} .

Let $\epsilon > 0$ and set $a_n = n^{-\alpha/2} (\log n)^2$ with α as before.

Lemma 5.

If (13) holds then

$$(19) \quad P(|R_{1n}| \geq \sigma_n a_n \epsilon) = O(a_n).$$

Proof:

Using Lemma 1 (iv) we have

$$P(|R_{1n}| \geq \sigma_n a_n \epsilon) \leq P(n^{1/2} |S_{n+1}^{-1} - 1| \geq \log n)$$

$$+ P(|L_n| \geq n^{(1-\alpha)/2} (\log n) \sigma_n \epsilon).$$

The first term is $O(n^{-1/2})$ by Lemma 2. The second term is bounded above by

$2(1 - \Phi(n^{(1-\alpha)/2} \epsilon \log n) + A_0 \Gamma_n)$, which is easily shown to be $O(a_n)$. This proves the lemma.

3.3 Convergence of R_n

For this we need the following conditions

1. H' satisfies a Lipschitz condition of order α (α as before) on $(0,1)$, i.e., for some finite constant A_1

$$(20) \quad |H'(x) - H'(y)| \leq A_1 |x - y|^\alpha \text{ for } 0 < x, y < 1.$$

2. There exists a finite number A_2 such that

$$(21) \quad n^{-1} \sum_j |c_{jn}| \leq A_2.$$

3. There exists a number $\tau > 0$ such that

$$(22) \quad \sigma_n \geq \tau \text{ for } n \text{ large enough.}$$

Remarks

1. In many cases we would expect (21) and (22) to hold; (21) because we will often have bounded $\{c_{jn}\}$ to guard against outliers ((21) is of course weaker than assuming bounded $\{c_{jn}\}$); (22) because often $\sigma_n^2 \rightarrow \sigma^2 > 0$. Condition (20) however places rather severe restrictions on H and in many cases might not hold.

2. Under (20), (21) and (22) a sufficient condition for (16) to hold is that

$$(23) \quad \max \{j/(n+1)(1 - j/(n+1)) c_{jn} H_{jn}' : j=1, \dots, n\delta \text{ and } j=n(1-\delta), \dots, n\} \\ \approx O(n^{(1-\alpha)/6})$$

for some $0 < \delta < 1/2$.

Lemma 6

Let $\epsilon > 0$ and a_n as in Lemma 5. If (20), (21) and (22) hold, then

$$(24) \quad P(|R_n| \geq \sigma_n a_n \epsilon) = O(n^{-1/2}).$$

Proof:

Using (20), (21) and (22), we get

$$P(|R_n| \geq \sigma_n a_n \epsilon) \leq P(n^{1/2} \max_j |V_{jn}| \geq \epsilon_0 (\log n)^{2/(1+\alpha)}),$$

where $\epsilon_0^{1+\alpha} = \epsilon \tau / A_1 A_2$. Using Lemma 3, the R.H.S. is $O(n^{-1/2})$, which yields the result.

We now state the final result. Let F_n denote the distribution function of $n^{1/2}(T_n - \mu_n)/\sigma_n$.

Theorem 1.

If (13), (20), (21) and (22) hold, then

$$(25) \quad \sup_{-\infty < x < \infty} |F_n(x) - \Phi(x)| = O(n^{-\alpha/2}(\log n)^2).$$

Proof:

Let a_n be as before and apply Lemma 1 (i) to get

$$\begin{aligned} F_n(x) &= P(L_n + R_n + R_{1n} \leq \sigma_n(x + a_n - a_n)) \\ &\leq P(L_n \leq \sigma_n(x + a_n)) + P(R_n \leq -a_n \sigma_n/2) + P(R_{1n} \leq -a_n \sigma_n/2). \end{aligned}$$

Using Lemmas 4, 5, and 6, we conclude

$$(26) \quad F_n(x) \leq \Phi(x) + r_n,$$

where $r_n = O(a_n)$ independent of x .

Using Lemmas 1 (ii), 4, 5, and 6, we also conclude that

$$(27) \quad F_n(x) \geq \Phi(x) - r_n$$

with r_n as before.

Combining (26) and (27) we get (25).

Remarks

1. The best rate we can obtain in (25) is $n^{-1/2}(\log n)^2$ which is the case when (20) holds with $\alpha = 1$. A sufficient condition for this is that H'' exists and is bounded on $(0,1)$.

2. By the same argument as above the rate in (25) can be improved to $O(n^{-\alpha/2}(\log n)^{1+d})$ with $d = 0$ if $\alpha < 1$ and $d > 0$ if $\alpha = 1$.

4. Special case

An important case is that in which the $\{c_{jn}\}$ are zero for j/n outside a fixed interval bounded away from 0 and 1. We state it as follows

$$(28) \quad c_{jn} = 0 \text{ for } j \notin \{n\lambda_1, \dots, n(1-\lambda_1)\} + 1$$

for some $0 < \lambda_1 < 1/2$.

An example of this is the trimmed mean with fixed trimming proportion λ_1 .

We also make the following assumption on H .

H'' exists and is bounded on some interval $[\lambda_2, 1-\lambda_2]$ with $0 < \lambda_2 < \lambda_1$; i.e., there exists a finite number A_4 such that

$$(29) \quad |H''(x)| \leq A_4 \text{ for all } x \in [\lambda_2, 1-\lambda_2].$$

Remark

In many cases H'' behaves badly only at 0 and 1, so (29) might not be unreasonable.

Theorem 2

If (21), (22), (28) and (29) are satisfied, then

$$(30) \quad \sup_x |F_n(x) - \Phi(x)| = O(n^{-1/2}(\log n)^2).$$

Proof

Start off as in Theorem 1. The terms with L_n and R_{1n} can be handled as before. Consider the term containing R_n . Let $\lambda > 0$ be some real number and define

$$V_n(\lambda) = \{ \max_j |V_{jn}| < \lambda \}.$$

Then for $\epsilon > 0$

$$\begin{aligned} P(|R_n| \geq \epsilon \sigma_n a_n) &\leq P(|R_n| \geq \epsilon \sigma_n a_n; V_n(\lambda)) + P(V_n(\lambda)^c) \\ &= p_{1n} + p_{2n} \quad (\text{say}) \end{aligned}$$

By Lemma 3, $p_{2n} = O(n^{-1/2})$. Convergence of p_{1n} follows from observing that on $V_n(\lambda)$ for λ chosen suitably small and n large enough, $\lambda_2 < U_{jn}^* < 1 - \lambda_2$ for $j = n\lambda_1, \dots, n(1-\lambda_1)$.

The theorem follows.

Example

Consider the trimmed mean with fixed trimming proportion λ_1 . In this case

$$(31) \quad T_n = ([n(1-\lambda_1)] - [n\lambda_1])^{-1} \sum_{j=[n\lambda_1]}^{[n(1-\lambda_1)]+1} X_{jn}$$

so $H = F^{-1}$ and $c_{jn} = ([n(1-\lambda_1)] - [n\lambda_1])^{-1}$, $j = [n\lambda_1], \dots, [n(1-\lambda_1)] + 1$

and zero otherwise. For many continuous distributions H' and H'' are bounded and continuous on any interval bounded away from 0 and 1, so (29) should hold. The other conditions are easily verified in this case and Theorem 2 applies.

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